Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Particular integral -Type 3 and Type 4

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Particular integral -Type 3 and Type 4

1 Particular integral -Type 3 and Type 4

Study focus: Particular integral -Type 3 and Type 4. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In previous session, we learned how to find the particular integral where f of x comma y equals sine e but x plus b y as well as sine of x plus b y or cos of x plus b y in this session.

Key details

- We will take up two more cases.
- That is finding Pi when f of x comma y is a polynomial in x and y.
- That is f of x comma y equals x power m times y power n.
- Also, we look at the general case.
- Or when f of x comma y is not is not a polynomial in x and y.
- That is when f of x comma y equals x power m dot y power n or at polynomial in x and y.
- How to how do we find out particular integral?
- As that the formula for particular integral is going to be one divided by f of d, d dash of f of f of x, y.
- As we have seen in ordinary differential equation, here also we bring f of b, comma d dash to the curator so that it's going to be z equals f of b, d dash inverse.
- X power m dot by power n.
- Then we need to expand f of b, d dash inverse as an infinite series, and then operate by x power m dot by power n.
- As a special case.
- How do you find out the particular integral when we have a polynomial on the right hand side?
- Square plus y square.
- Either we can take out d square common or d square out common because m and n are equal here.

- In case we take out d square common, what happens?
- Because in the polynomial case, so it is convenient to have either one plus something or one minus something before we apply binomial series.

Example or application from the lecture

- Let's start with case number three.
- suppose f of d gamma d dash doesn't contain the constant term that is the term which is independent of d and d dash then we have two cases we expand f of d gamma d dash inverse power minus one in power of d dash inverse dot d if m is less than n or in power sub d inverse dash dot d dash if m is greater than n.
- Okay, let's take an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

minus square and d da square which is nothing but d square of one minus d da square divided by d square so this is going to be d na square.

Key details

- And you can easily see that this is like one minus x or one minus something that something is this b inverse square dot b dash square.
- Moreover, all these things are in the denominator.
- So in the next time we need to bring it up.
- D square minus theta square inverse of x square plus y square.
- So when we use the series, so that's going to be the minus two of one minus the inverse square then d dash square of x square plus y square because m and n are equal to two so we stop the series so when we reach d square as well as d data square.
- So because the highest power of m and n is two, we stop the series at t square and d ta square.
- So that is why if we look at the formula for binomial series, these terms are not considered because anyway.
- D inverse power four and so on of this x square plus y square become zero.
- So when you write the formula we are considering only one as well as x because x is t square and d naught square and d da square.
- So it means we are getting power minus two of one plus.
- B inverse square.
- Then we have our d dash square.
- Other terms are neglected.
- Outside we have x square plus y square.
- Then we should take it inside.
- D inverse square dot d dash square of x square plus y square.
- Next step we just read an as it is.
- So d dash square means we need to find out the partial derivative of this with respect to y.

- Then what is t inverse square of 2?
- So first time this is 2x, second time this is 2x square by 2, but cancellation 2 on 2 we get x square.
- Then what is d power minus two?
- plus x square dot y square by two.
- So as we know from the earlier concept, because m and n are equal, we have two choices.
- Either we can take d square common or we can take d dash square common.
- for a PD there are chances of getting different answers.

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EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 10:00–14:59

Okay, let's take a another example.

Key details

- Find the particular integral for the partial differential equation.
- Either we can take a t outside or t dash outside.
- Assume we take D outside, so that leads to this particular step, and we get one plus x, then that series formula minus two.
- Because the highest power of x m and n or the values of m and n r equals one, it is enough to go up to d and d dash.
- two d dash and d power minus one in it these things are zeros so once we take three x plus two y inside so that leads to three x plus two y and two d dash d power minus one three x plus two y.
- As we know, this is nothing but integration of x with respect to x, integration with respect to x, keeping y constant.
- So 3x becomes 3x square by 2, so y becomes 2xy because 2y is constant and dx become x.
- Then what is d dash of this part?
- Of course partial derivative.
- Outside we have one more 2, therefore we get minus 4x.
- We need to integrate with respect to x two times.
- plus x square y.
- In case we take out the d dash common, so that leads to this particular answer.
- You can clearly see that.
- Both answers are different and it is valid also.
- If you remember when we discuss case number two, it was said that when f of minus d square minus d dash and minus d dash square become zero.
- So this is a d square because we replace d square by minus a square, so it is a d square minus a d dash and d dash square zero.
- How do you find out P PI?
- So even that case can be solved by using this method.
- Assume that f of b, d dash is reducible.

- As from the beginning of this chapter, so if we can express f of b comma d dash to be something like AI.
- D plus BI D dash plus CI.
- If F of B gamma D dash can be reduced to this form, then we say it is reducible.
- Then how do we apply the procedure here?

Example or application from the lecture

- When a form x comma y is not any of the forms as given in the above cases or.

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Core explanation · 15:00–19:59

Remember in this case C is taken zero.

Key details

- A1P and D dash Z is nothing but douz by dou y which is same as Q so therefore we get B1 Q equals F of X com Y.
- If you remember Lagrange's equation.
- is exactly the same so what is Lagrangius linear equation capital p of x y z times all p capital q of x y z small q equals r so it we learned that this is how lagrangius linear equation looks like so here also we have the same thing.
- First and second fractions and first and third fractions.
- So from the first and second ratios, we can get when we have variable separable that is b1 dx equals a one dy.
- After integration that leads to B1 minus A1 to Y equal to C where A C is arbitrary constant.
- Remember in later stage we need to solve for y.
- Okay, when we take the first and third fractions or first and third ratios.
- We get dx by a1 equals dz by f of x comma y.
- That is a1 dz equals f of x karma b1x minus c by a1 dot bx.
- Since we are looking for a particular integral, we need to put C1 to be zero.
- F of x comma b1 x minus a1.
- This procedure is repeated based on how many factors are involved in the problem.
- Also we looked at the general.
- That is when Mf of X comma y is not e pous b y not sine of x plus b y or cos of x plus b y or not a polynomial in x and y.

Example or application from the lecture

- Let's see.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral -Type 3 and Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00050	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...
SRC00052	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare...

C Revision and retrieval practice